

4D Decomposition

Overlapping Rules → Disjoint Cells → TRF

Input LRF (first-match order)

r1: TCP 10.0.0.x / 80–1023 → DENY

r2: IP ANY → ALLOW (r2 generalizes r1)

↓
sweep-line 3D

After 4D Decomposition (disjoint cells)

(UDP and ICMP: one ALLOW cell each, inherited from r2)

TCP · src ≠ 10.0.0.x

port 0–65535

→ **ALLOW**

TCP · 10.0.0.x

port 0–79

→ **ALLOW**

TCP · 10.0.0.x

port 80–1023

→ **DENY**

TCP · 10.0.0.x

port 1024–65535

→ **ALLOW**

↓
directional merge

TRF Output (TCP subtree, Ordering 4)

[protocol]=TCP → [dst_ip]=ANY → [dst_port] → [src_ip]

port 0–79 → ALLOW

port 1024–65535 → ALLOW

port 80–1023 → [src_ip]: 10.0.0.x → DENY, other → ALLOW

Every packet reaches exactly one leaf; the DENY region equals $M(r1)$, as first match requires.